Homework - New chat

L6_S1_A.pdf
PDF

I want you to generate a lecture scribe for the PDF lecture slides ive uploaded. while youre at it, be very precise with the steps and reasonably mention any assumptions, proofs, definitions and examples solutions exactly as presented in the provided lecture slides. Make a proper logical flow so that any student can follow the reasoning at the time of revision. DO NOT INTRODUCE ANY NEW MATERIAL THAT IS NOT MENTIONED IN THE UPLOADED DOCUMENT.

Got it — thanks for being very explicit. I'll stay **strictly faithful** to the uploaded slides, introduce **no new material**, and structure this as a **clean, revision-ready lecture scribe** with precise logic, definitions, assumptions, derivations, and worked examples exactly as presented.

Below is a **Lecture Scribe for Lecture 6 (Discrete Random Variables, Expectation, and Problem Solving)** based **only** on the uploaded slides .

CSE400 – Fundamentals of Probability in Computing

Lecture 6: Discrete Random Variables, Expectation and Problem Solving

Instructor: Dhaval Patel, PhD
Date: January 22, 2025

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1. Random Variables: Motivation and Concept

Definition

A **random variable (RV)** is a function that maps outcomes from a **sample space** to **numerical values**.

- Instead of working directly with outcomes (e.g., coin toss sequences), we work with numbers derived from them.
- This allows us to assign probabilities to numerical values.

Visualization (as per slides)

- The **distribution of a random variable** can be visualized using a **bar diagram**.
- The **x-axis** represents the possible values of the random variable.
- The **height of the bar** at value a_i represents:

$$\Pr[X = a_i]$$

- Each probability is computed from the probability of the **corresponding event in the sample space**.

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2. Types of Random Variables

2.1 Discrete Random Variables

A random variable is **discrete** if:

- It has **countable support**
- Probabilities are assigned to **single values**
- Each possible value has **strictly positive probability**
- It is described using a **Probability Mass Function (PMF)**

2.2 Continuous Random Variables (for contrast only)

- Have **uncountable support**
- Described using a **Probability Density Function (PDF)**
- Probabilities are assigned to **intervals**
- Each exact value has **zero probability**

(Note: No continuous RV calculations are performed in this lecture.)

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3. Example 1: Tossing 3 Fair Coins

Experiment Description

- Toss **3 fair coins**
- Let random variable **Y** = number of heads observed

Possible Values of Y

$$Y \in \{0, 1, 2, 3\}$$

Probability Calculations

- $P(Y = 0) = P(t, t, t) = \frac{1}{8}$
- $P(Y = 1) = P(t, t, h) + P(t, h, t) + P(h, t, t) = \frac{3}{8}$
- $P(Y = 2) = P(t, h, h) + P(h, t, h) + P(h, h, t) = \frac{3}{8}$
- $P(Y = 3) = P(h, h, h) = \frac{1}{8}$

Probability Distribution Summary

$$P(Y = 0) = \frac{1}{8}, \quad P(Y = 1) = \frac{3}{8}, \quad P(Y = 2) = \frac{3}{8}, \quad P(Y = 3) = \frac{1}{8}$$

Total Probability Check (Assumption: RV must take exactly one value)

$$1 = P\left(\bigcup_{i=0}^{3}\{Y = i\}\right) = \sum_{i=0}^{3} P(Y = i)$$

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4. Probability Mass Function (PMF)

Definition

A random variable that can take **at most a countable number of values** is called **discrete**.

Let:

- X be a discrete random variable
- Range (possible values):

$$R_X = \{x_1, x_2, x_3, \dots\} \quad \text{(finite or countably infinite)}$$

PMF Definition

The function:

$$P_X(x_k) = P(X = x_k), \quad k = 1, 2, 3, \dots$$

is called the **Probability Mass Function (PMF)** of X.

Fundamental Property of PMF

Since X must take **one of the possible values**:

$$\sum_{k=1}^{\infty} P_X(x_k) = 1$$

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5. PMF Example: Given Functional Form

Problem Statement

The PMF of a random variable X is:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

where $\lambda > 0$.

Find:

1. $P(X = 0)$
2. $P(X > 2)$

Step 1: Find the Constant c

Using the PMF property:

$$\sum_{i=0}^{\infty} p(i) = 1$$

Substitute:

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using the known series expansion:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

Thus:

$$ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

Step 2: Compute $P(X = 0)$

$$P(X = 0) = e^{-\lambda}\frac{\lambda^0}{0!} = e^{-\lambda}$$

Step 3: Compute $P(X > 2)$

Using complement:

$$P(X > 2) = 1 - P(X \leqslant 2)$$

$$= 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

$$= 1 - (e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2}e^{-\lambda})$$

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6. Bayes' Theorem (Recap)

Starting Point

Using:

$$P(A \cap B) = P(B \mid A)P(A)$$

We obtain:

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_j P(A \mid B_j)P(B_j)}$$

This is called **Bayes' Formula (Proposition 3.1)**.

Definitions

- **Prior probability**:
  $P(B_i)$ — formed before observing data
- **Posterior probability**:
  $P(B_i \mid A)$ — computed after observing event A

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7. Bayes' Theorem – Example 1: Auditorium with 30 Rows

Setup

- Auditorium has **30 rows**
- Row 1 has **11 seats**
- Row 2 has **12 seats**
- ...
- Row 30 has **40 seats**

Selection Process

1. A row is selected **uniformly at random**
2. A seat is selected **uniformly at random within that row**

Questions

1. Compute $P(\text{Seat 15})$
2. Compute $P(\text{Row 20} \mid \text{Seat 15})$
3. Compute $P(\text{Row 20})$

Key Formula Used

$$P(S_{15}) = \sum_{k=5}^{30} P(S_{15} \mid R_k) P(R_k)$$

Using:

- $P(R_k) = \frac{1}{30}$
- $P(S_{15} \mid R_k) = \frac{1}{\text{number of seats in row } k}$

Final numerical value (as shown in slides):

$$P(S_{15}) = 0.0342$$

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8. Bayes' Theorem – Example 2: Communication System (Receiver)

System Description

- Binary data $\{0, 1\}$ is transmitted
- Receiver (RX) may make errors:
  - 0 may be detected as 1
  - 1 may be detected as 0

Given

- A set of **conditional probabilities** describing detection errors

(The lecture slide introduces the setup; detailed numerical evaluation continues beyond this slide.)

End of Lecture Scribe

Final Notes (for Revision)

- Every PMF **must sum to 1**
- Bayes' theorem **reverses conditioning**
- All examples strictly follow:
  - Proper normalization
  - Complement rule
  - Law of total probability

If you want, next I can:

- Convert this into **exam-ready handwritten notes format**
- Generate a **one-page revision cheat sheet**
- Cross-check this against **your midterm syllabus**

Just say the word.